

# Balanced Similarities for Directed Packet Couplings

## Rank Rescue and the Original-Norm Conditioning Price

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### Abstract

The two-sided Schur product of RH-163 is the sharp feedback quantity, but the single-norm RH-161 theorem remains useful when an actual block norm can be optimized. For  $A = \begin{pmatrix} A_P & B \\ C & A_Q \end{pmatrix}$ , conjugation by  $D_t = \text{diag}(tI_P, I_Q)$  replaces the directed couplings by  $tB$  and  $t^{-1}C$ . We prove

$$\min_{t>0} \max(tb, c/t) = \sqrt{bc}, \quad t_* = \sqrt{c/b},$$

where  $b = \|B\|$  and  $c = \|C\|$ . Hence the balanced RH-161 rank gate is

$$\max(a, d)\sqrt{bc} < 1.$$

The similarity does not come for free: transforming the Riesz projector back to the original Hilbert norm multiplies its error bound by  $\max(t_*, t_*^{-1})$ . Thus an imbalanced system may have stable spectral rank but a poorly conditioned packet graph. The triangular limit is treated directly because the optimizing similarity becomes singular. A 512-case audit verifies the exact off-diagonal norm and optimum. No physical all-level balance is claimed.

## 1 The scalar block-similarity family

Let  $P$  be an orthogonal projection,  $Q = I - P$ , and write

$$A = \begin{pmatrix} A_P & B \\ C & A_Q \end{pmatrix}.$$

For  $t > 0$  put

$$D_t = \begin{pmatrix} tI_P & 0 \\ 0 & I_Q \end{pmatrix}, \quad \tilde{A}_t = D_t A D_t^{-1} = \begin{pmatrix} A_P & tB \\ t^{-1}C & A_Q \end{pmatrix}.$$

The block diagonal part and its resolvent are unchanged. The off-diagonal part  $\tilde{E}_t$  satisfies the exact identity

$$\|\tilde{E}_t\| = \max\{t\|B\|, t^{-1}\|C\|\}.$$

Indeed,  $\tilde{E}_t^* \tilde{E}_t$  is block diagonal with diagonal blocks  $t^{-2}C^*C$  and  $t^2B^*B$ .

## 2 Optimal balance

**Theorem 1** (Exact scalar balance). *Let  $b, c > 0$ . Then*

$$\inf_{t>0} \max(tb, c/t) = \sqrt{bc},$$

*and the unique minimizer is  $t_* = \sqrt{c/b}$ .*

*Proof.* For every  $t$ , the maximum is at least the geometric mean  $\sqrt{(tb)(c/t)} = \sqrt{bc}$ . At  $t_*$  the two arguments are equal to that value, proving optimality and uniqueness.  $\square$

The Schur feedback  $adbc$  is unchanged by this similarity. Balancing is not a new spectral invariant; it is the optimal way to feed the same directed data into a one-norm Neumann theorem.

## 3 Balanced Neumann certificate

Let  $\Gamma$  isolate the packet block from the complement and suppose

$$\sup_{z \in \Gamma} \|(z - A_P)^{-1}\| \leq a, \text{ and } \sup_{z \in \Gamma} \|(z - A_Q)^{-1}\| \leq d.$$

Put  $M = \max(a, d)$  and let  $L = |\Gamma|$ .

**Theorem 2** (Balanced rank and projector bounds). *If*

$$\theta = M\sqrt{bc} < 1,$$

*then  $\Gamma$  remains in the resolvent under the full off-diagonal homotopy, and the enclosed Riesz rank equals  $\text{rank } P$ . In the balanced norm,*

$$\|\tilde{\Pi} - P\| \leq \delta_t := \frac{L}{2\pi} \frac{M^2 \sqrt{bc}}{1 - M\sqrt{bc}}.$$

*In the original norm,*

$$\|\Pi - P\| \leq \max(t_*, t_*^{-1}) \delta_t.$$

*If the last quantity is below one, it supplies a graph certificate.*

*Proof.* Apply the RH-161 Neumann theorem to  $\tilde{A}_{t_*}$ , whose off-diagonal norm is  $\sqrt{bc}$  by Theorem 1 [Program, 2026a]. Similar operators have the same spectrum and Riesz rank. Since  $D_t$  commutes with  $P$ ,

$$\tilde{\Pi} - P = D_t(\Pi - P)D_t^{-1}.$$

Thus  $\|\Pi - P\| \leq \|D_t^{-1}\| \|D_t\| \|\tilde{\Pi} - P\|$ , and  $\|D_t^{-1}\| \|D_t\| = \max(t, t^{-1})$ .  $\square$

Because  $ad \leq M^2$ , the Schur gate  $adbc < 1$  from RH-163 is never weaker than the squared balanced gate  $M^2bc < 1$  [Program, 2026b]. The value of the balanced theorem is its direct compatibility with single-norm perturbation machinery and its transparent conditioning ledger.

## 4 Triangular systems and graph asymmetry

If  $c = 0$  or  $b = 0$ , the infimum in Theorem 1 is zero but is not attained by an invertible  $D_t$ . One must not insert  $t = 0$  or  $t = \infty$  as a similarity. Instead, the original block operator is triangular and the enclosed rank is preserved exactly under spectral separation.

If  $c = 0$  but  $b$  is large, the packet range is invariant although the Riesz projection can have a large complementary-to-packet off-diagonal block. This explains the divergent similarity condition number: rank needs closed feedback, while a well-conditioned two-sided projection needs control of both directions.

## 5 The full feasible scaling interval

Balancing is the center of an explicit interval, not merely one lucky scale.

**Proposition 3** (Neumann-feasible similarities). *For  $b, c, M > 0$ , the transformed RH-161 gate*

$$M \max(tb, c/t) < 1$$

*holds exactly for*

$$Mc < t < \frac{1}{Mb}.$$

*This interval is nonempty if and only if  $M^2bc < 1$ . The balanced scale  $t_* = \sqrt{c/b}$  is its logarithmic midpoint and maximizes the smaller logarithmic distance to its two endpoints.*

*Proof.* The maximum is below  $M^{-1}$  precisely when both  $tb < M^{-1}$  and  $c/t < M^{-1}$ , giving the interval. It is nonempty exactly when  $Mc < (Mb)^{-1}$ . Taking logarithms shows that the midpoint is

$$\frac{1}{2}(\log(Mc) + \log((Mb)^{-1})) = \frac{1}{2}\log(c/b) = \log t_*.$$

□

The interval is useful for graph optimization. One may move away from  $t_*$  to reduce  $\max(t, t^{-1})$  while remaining inside the rank-feasible region. The optimal graph scale therefore need not be the optimal coupling scale; it is a one-dimensional certified minimization over this explicit interval.

## 6 Finite audit and route consequence

The audit draws 512 rectangular pairs  $(B, C)$ , computes the exact norm of  $\begin{pmatrix} 0 & tB \\ t^{-1}C & 0 \end{pmatrix}$ , and compares it with  $\max(tb, c/t)$ . It also tests the analytic minimizer against four neighboring scales. No failure occurs.

RH-164 supplies an optimized fallback certificate and a warning: a rescued rank gate is not automatically a usable packet graph. The next geometric task is to choose  $\Gamma$  so that  $a$  and  $d$  are controlled rather than postulated. No physical realization, contour, all-level margin, Gate A, self-adjoint operator, zeta identity, or RH claim is made.

## References

Prime Dynamics Theory Program. Packet-to-riesz relative-determinant assembly, 2026a. RH-161 manuscript.

Prime Dynamics Theory Program. A two-sided schur certificate for packet-to-riesz lifting, 2026b. RH-163 manuscript.